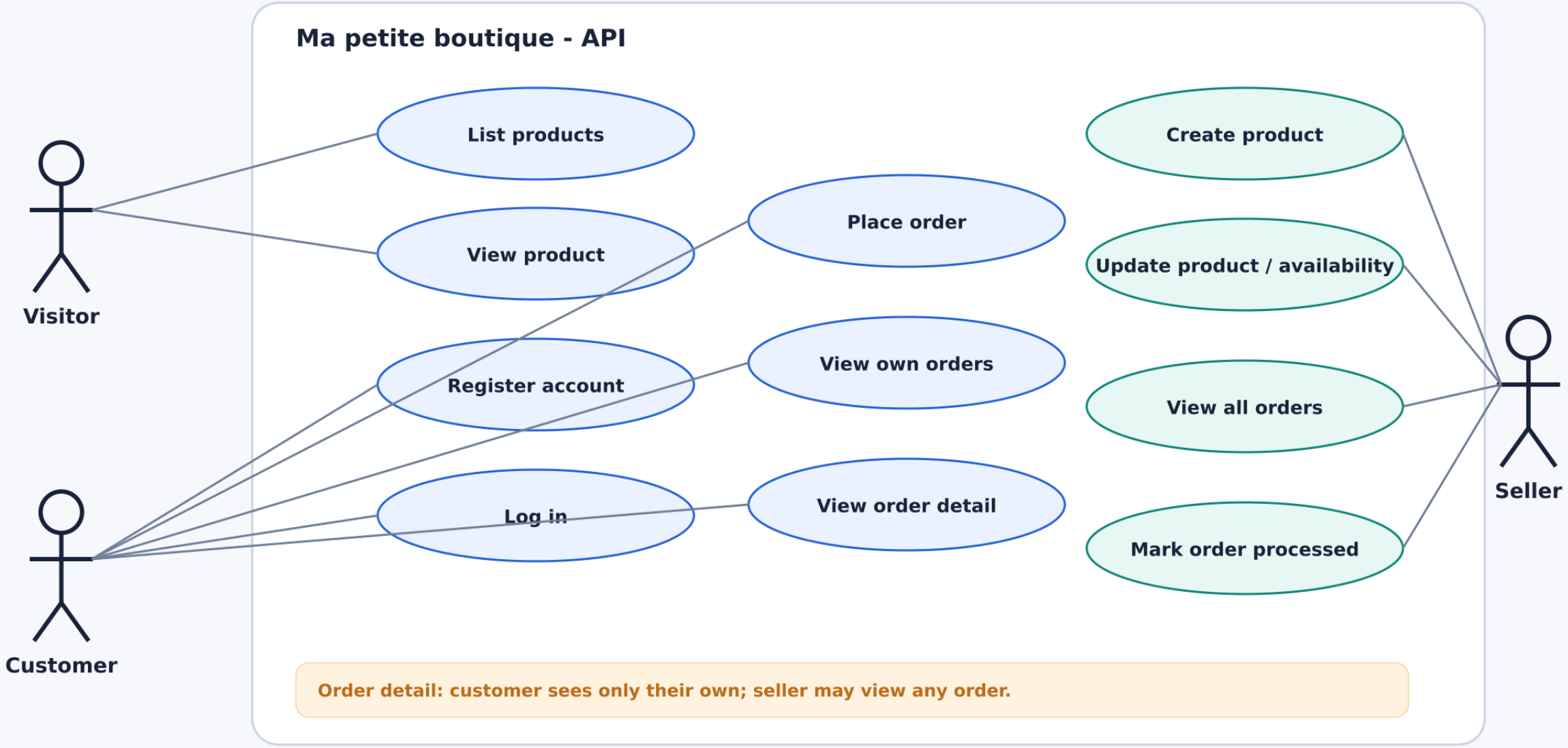
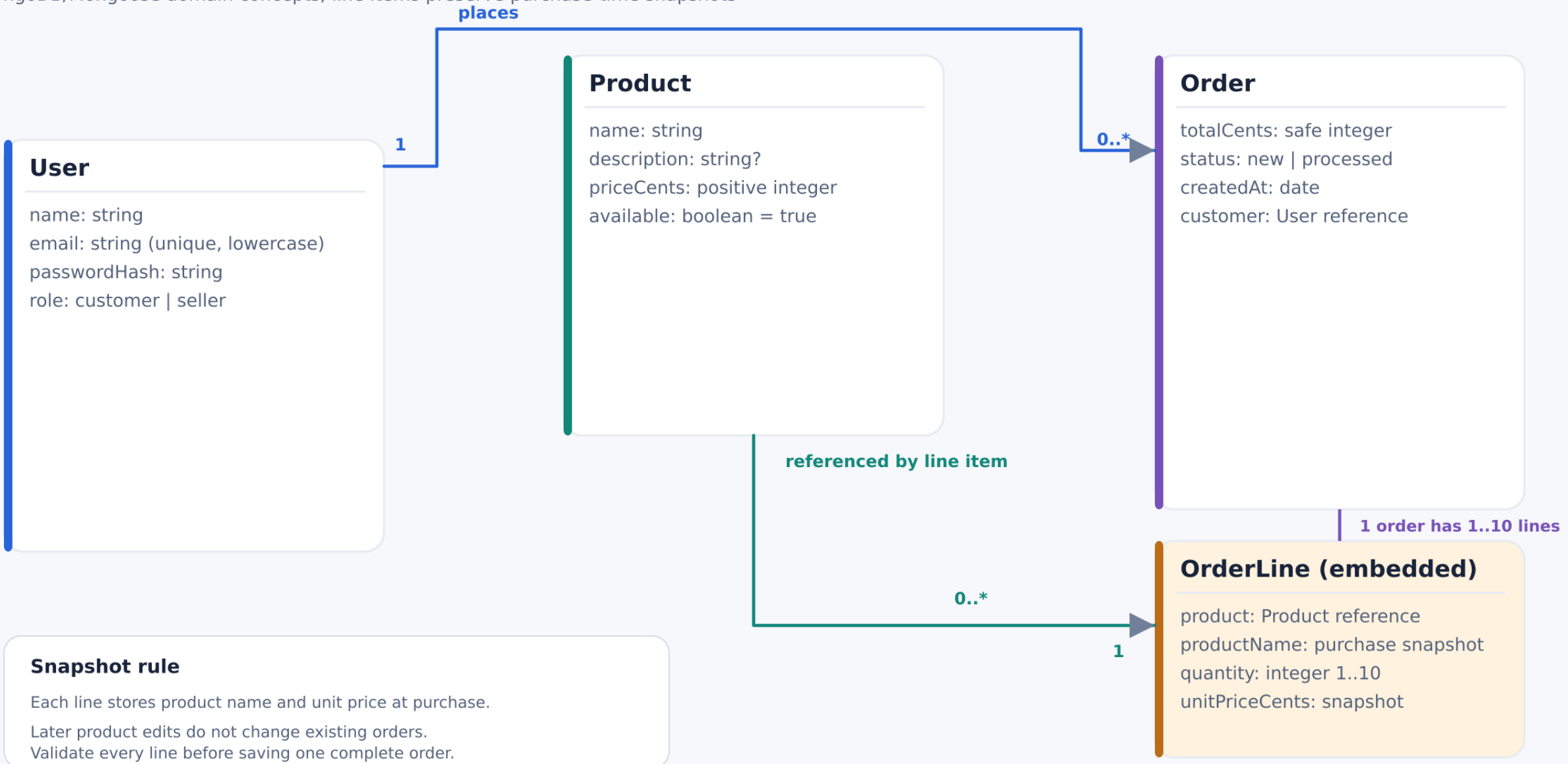
Use-case diagram

Actor permissions and user goals stated in the brief



Domain class diagram

MongoDB/Mongoose domain concepts; line items preserve purchase-time snapshots



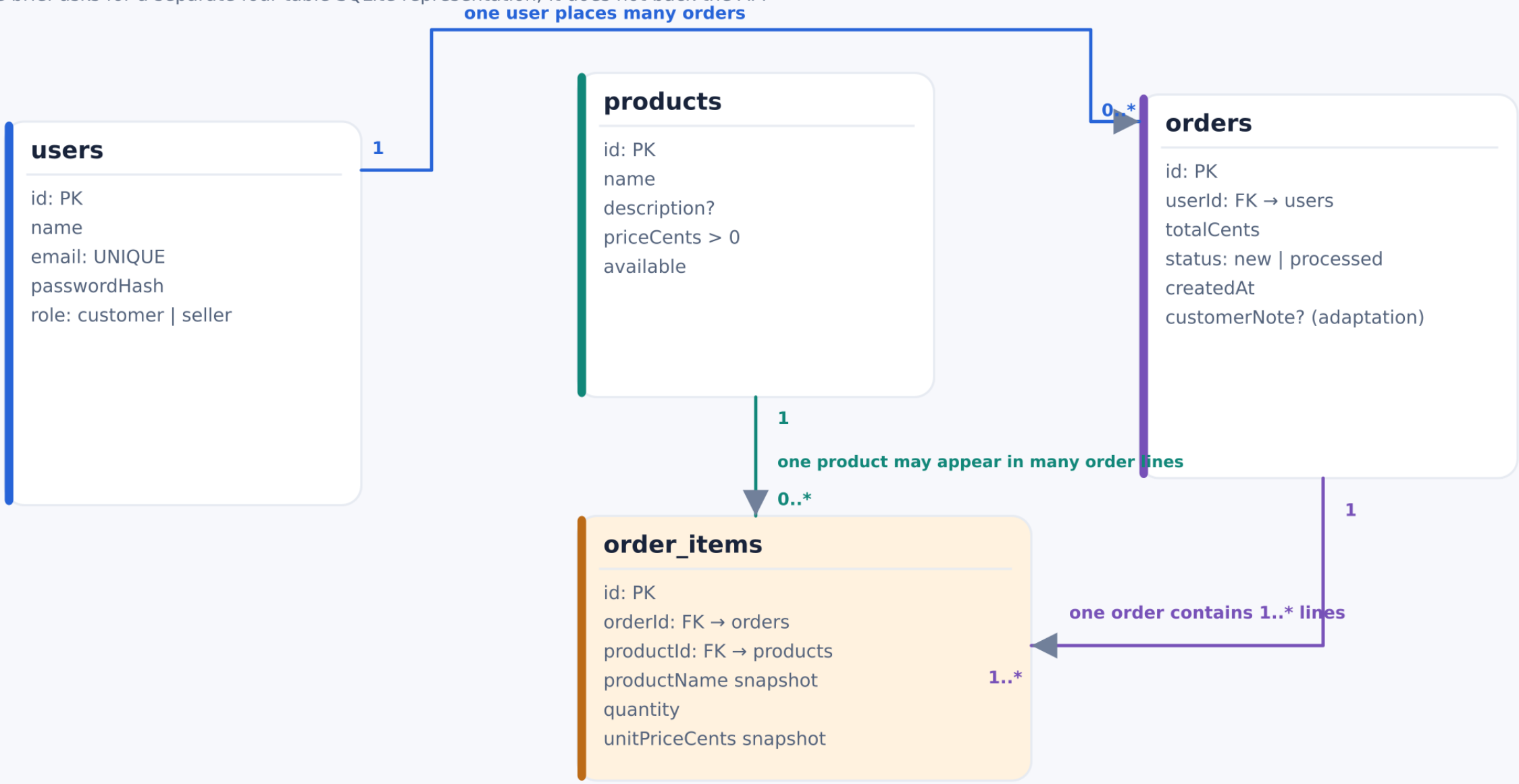
Snapshot rule

Each line stores product name and unit price at purchase.

Later product edits do not change existing orders.
Validate every line before saving one complete order.

Relational model (SQLite exercise)

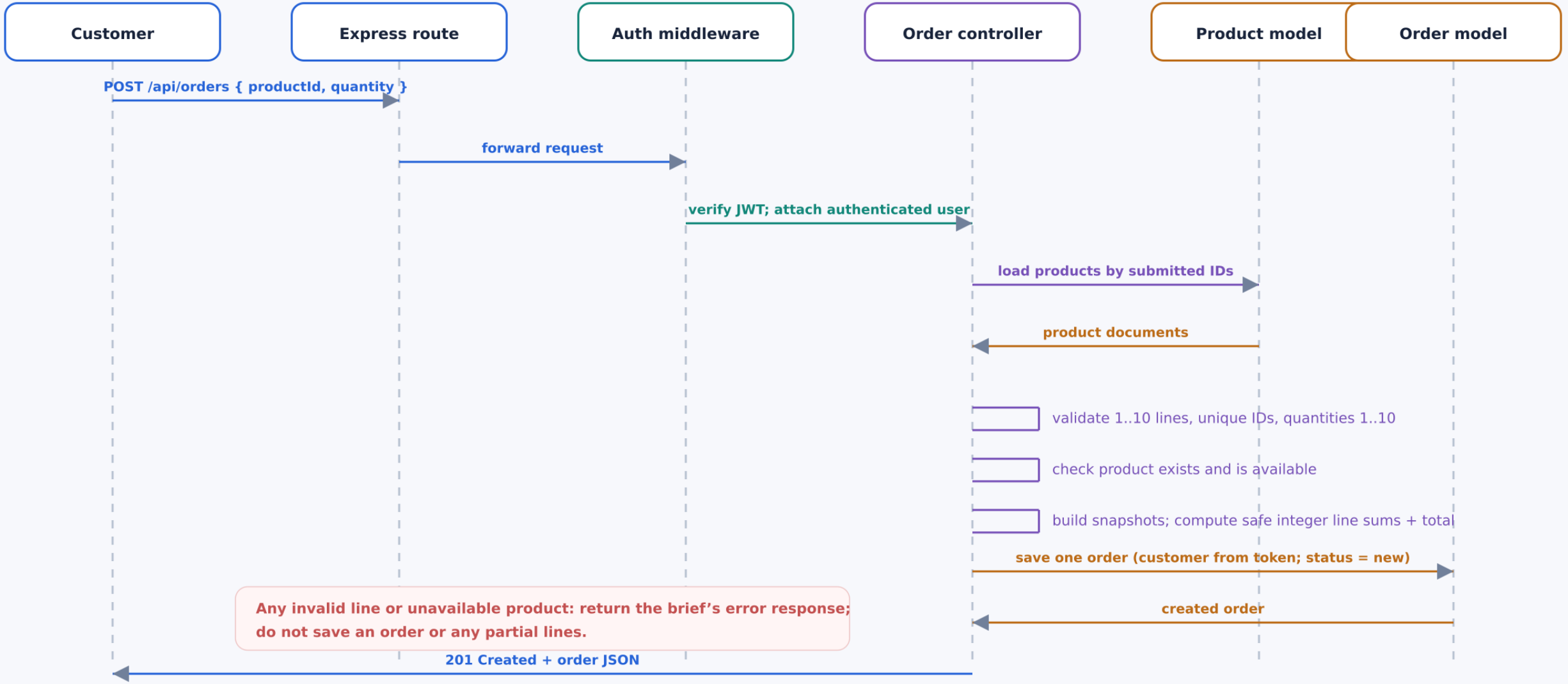
The brief asks for a separate four-table SQLite representation; it does not back the API



Primary/foreign keys and CHECK constraints protect the separate SQLite model; snapshot fields retain purchase values.

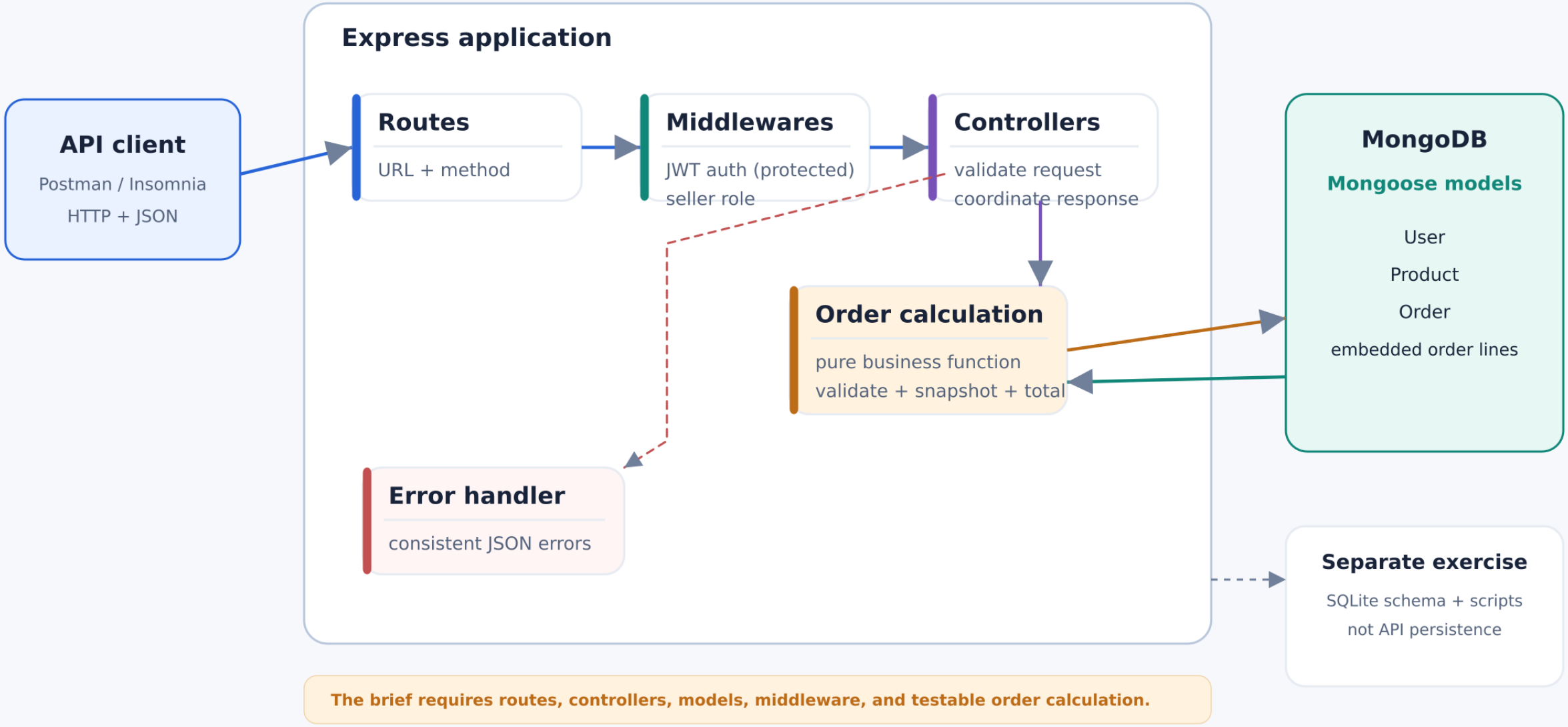
Sequence: place an order

Order flow and validation boundaries; only accepted orders are persisted



Recommended API architecture

Required responsibility split from the brief, with one explicitly isolated order-calculation function



Sequence: process an order

Seller-only status transition; a processed order cannot return to new

